

TIANHAO CAO

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OBJECTIVE

M.S. Data Science candidate at the University of British Columbia (Computational Linguistics) with hands-on experience across applied machine learning, NLP, LLM and retrieval-augmented generation systems, causal inference, and AI privacy research. Builds complete systems end to end, from data ingestion and transformer fine-tuning through hybrid retrieval, guardrails, and LLM-as-judge evaluation, with measured results on every project. Seeking applied machine learning, NLP, or data science work where rigorous experimentation and production-minded engineering shape real decisions.

EDUCATION

University of British Columbia	Vancouver, BC
M.S. in Data Science (Computational Linguistics)	Sep 2025 – Jun 2026

University of California, Santa Cruz	Santa Cruz, CA
B.A. in Business Management Economics	Sep 2019 – Jun 2022

TECHNICAL SKILLS

Programming	Python (primary), R, SQL, MongoDB, Bash
ML & AI	PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, Fine-Tuning, SHAP
LLM & GenAI	RAG Pipelines, Prompt Engineering, LLM-as-Judge Evaluation, Ragas, Guardrails
Retrieval & NLP	ChromaDB, BM25s, RRF, SentenceTransformers, spaCy, jieba, Whoosh, LangChain
Web & API	FastAPI, REST API Design, Pydantic, Gradio
Data Engineering	ETL Pipelines, Feature Engineering, Pandas, NumPy, Apache Parquet
MLOps & Cloud	Docker, AWS, Git/GitHub, CI/CD, Weights & Biases (W&B), Linux
Languages	English (Full Professional) Mandarin (Native)
Work Auth.	Canadian Post-Graduation Work Permit (PGWP) Eligible

EXPERIENCE

Applied Machine Learning Engineer	May 2026 – Jun 2026
<i>Internship</i>	<i>ednoda.com</i>

- Owned the sentence generation pipeline end to end, replacing LLM API calls with a fine-tuned RoBERTa masked-LM head that fills grammar-aware slots for a 76.1% judge-rated Good Rate at zero inference cost.
- Designed a hybrid RAG collection-discovery search over the textbook/unit catalog, fusing ChromaDB dense with BM25s sparse results via Reciprocal Rank Fusion (k=60) to lift recall from 0.634 to 0.994.
- Enforced guardrails on generated content, blocklisting existing, rejected, and previously suggested nodes under a lemma-aware filter for 0 exclusion violations across all 70 candidates.

PROJECTS

Membership Inference Attack on Fine-Tuned LLMs	UBC
<i>LLM Privacy Research, Kaggle Competition</i>	<i>GitHub Repository</i>

- Reached AUC 0.907 and TPR@FPR=0.1 of 0.697 on the validation split, a +55% AUC and +349% TPR improvement over the raw-loss baseline that ranked as the top-performing method in the competition.
- Designed an Improved Neighborhood Attack fusing dual-model features across four models (fine-tuned SmolLM2 135M/360M and their base checkpoints), capturing memorization signals invisible to single-model baselines.
- Engineered a 16-dimensional feature vector spanning fine-tuned losses, base losses, reference ratios, perturbation gaps, and cross-model agreement, feeding a 300-estimator Gradient Boosting classifier.
- Replaced slow BERT mask-and-fill neighbor generation with a fast token-drop perturbation strategy (10% dropout, 5 neighbors) and traced 86% of feature importance to cross-model reference signals.

MDS-CL RAG Assistant

Hybrid Retrieval and Ragas Evaluation

Independent
GitHub Repository

- Built an end-to-end domain RAG assistant grounded in a scraped program website, chunked with a two-stage Markdown-header plus 800-character recursive splitter into a ChromaDB dense store and a BM25s sparse index.
- Fused dense and sparse results with Reciprocal Rank Fusion (k=60) to improve recall over either method alone, then constrained Gemini 2.5 Flash generation to retrieved context under a strict system prompt.
- Added an intent-classification layer with per-intent few-shot examples that routes question types and short-circuits out-of-scope queries before retrieval, and scored the pipeline with Ragas on four metrics.
- Shipped a Gradio chat interface with streaming output and a settings sidebar for live API key, endpoint, model, and Top-K reconfiguration without restarting the server.

Full-Stack Corpus Search Engine

Bilingual Search with ML Annotation (4-person team)

UBC
GitHub Repository

- Co-built a bilingual (English and Mandarin) full-text search engine on FastAPI and Whoosh, where a custom jieba-based ChineseAnalyzer delivered accurate segmentation and sub-second multi-language queries.
- Designed 7 Pydantic-validated REST endpoints for bilingual multi-field search, runtime JSON/JSONL corpus upload that updates the index without a server restart, label statistics, and stopword-filtered word frequency.
- Owned a 3-stage annotation pipeline: bilingual TF-IDF plus LinearSVC pseudo-labeling, active learning prioritization via Query-by-Uncertainty and Query-by-Committee, and few-shot data augmentation.
- Compared neural transfer-learning freezing strategies and a multi-task variant jointly optimizing fraud classification with silver NER, then containerized the application with Docker for reproducible deployment.

LLM Response Preference Classification

Kaggle: Chatbot Arena Human Preference

Independent
Code Available on Request

- Fine-tuned microsoft/deberta-v3-base as a 3-way cross-encoder on 57K Chatbot Arena head-to-head pairs, reaching val log loss 1.049 and becoming the first model to beat the 1.097 class-prior baseline.
- Diagnosed a TF-IDF plus Logistic Regression baseline (1.129) that underperformed a constant-prior guess as a feature-representation ceiling rather than a tuning problem, which justified moving to a cross-encoder.
- Designed swap augmentation that duplicates every pair with response order exchanged and the label flipped, doubling training data to 115K rows and structurally removing the positional bias a cross-encoder would otherwise learn.
- Trained with fp16 mixed precision, 10% warmup plus cosine decay, gradient accumulation, and load_best_model_at_end, scaling from 1 to 3 epochs after confirming eval loss was still descending.

Membership Inference on Vision-Language Models

Multimodal Privacy Research

Independent
GitHub Repository

- Raised VLM membership inference from a raw-loss baseline (AUC 0.50) to AUC 0.849 and TPR@FPR=0.1 of 0.388 by fusing loss-ratio with multimodal perturbation features via Logistic Regression.
- Adapted the M⁴I framework (Hu et al., NeurIPS 2022) to a CLIP ViT-B/32 plus SmolVLM-256M-Instruct stack, replacing legacy ResNet-152/LSTM shadow models with an off-the-shelf cross-modal extractor.
- Extended MIA to the multimodal neighborhood setting with text dropout plus image Gaussian noise, patch masking, and random crop, extracting 17 cross-modal sensitivity features across five attack variants.
- Identified that CLIP image-text cosine similarity dominates the membership signal at 48% of XGBoost feature importance, shifting the attack paradigm from model behavior to data alignment.

BERT Encoder Implemented from Scratch

Transformer Internals in Raw PyTorch

Independent
Code Available on Request

- Implemented multi-head self-attention from first principles (per-head Q/K/V projections, scaled dot-product attention, head concatenation, output projection) plus a single-head module to verify the math.
- Assembled the full encoder by hand: position-wise feed-forward blocks, post-LayerNorm residual connections, summed token/positional/segment embeddings, and an MLM head forming BERTForMLM.

- Wrote the masked language modeling objective from scratch (15% token selection, the 80/10/10 mask/random/unchanged split, -100 ignore indexing) and pre-trained on WikiText-2 with a custom Dataset and DataLoader.
- Drove average MLM loss from 7.71 to 6.73 to 6.65 over 3 epochs in a CPU-feasible configuration and validated masked-token inference with coherent Top-5 vocabulary predictions.

Betty Talker: Recipe RAG Pipeline

Retrieval-Augmented Recipe Assistant

UBC

GitHub Repository

- Built a full RAG pipeline from scratch, parsing a public-domain cookbook into structured JSON with regex extraction and encoding recipes with all-MiniLM-L6-v2 embeddings for cosine-similarity retrieval.
- Designed a few-shot LLM intent classifier routing queries into recipe search, ingredient scaling, or off-topic refusal, keeping every response grounded strictly in retrieved context.
- Shipped a scaling module that handles Unicode fractions and computes scale factors from a direct multiplier or target serving size, plus a CLI, .env key management, and one-command setup.

Yelp Review Sentiment Analysis

Modular NLP Deep Learning Pipeline

Independent

GitHub Repository

- Replaced a monolithic notebook with a modular pipeline (data_loader, eda, preprocessing, models, train) for 5-class star-rating classification, reaching best validation Macro F1 of 0.557 on 27,500 Yelp reviews.
- Benchmarked three model families: a TF-IDF (unigram plus bigram) Logistic Regression baseline, a CBOW classifier over 300-dimensional pretrained spaCy vectors, and a 3-layer Bidirectional LSTM.
- Implemented a generic training loop with F1-based early stopping (patience=3), checkpoint saving at the best epoch, and per-epoch train and validation loss reporting.

Credit Card Default Analysis

End-to-End Machine Learning Pipeline

UBC

GitHub Repository

- Selected an optimized XGBoost model for the best Precision/Recall balance (F1-score 0.55) on a heavily imbalanced 30,000-sample UCI credit dataset after benchmarking Logistic Regression and Random Forest.
- Ran extensive EDA on data distributions and class imbalance, then applied resampling and feature scaling to improve model robustness across the end-to-end pipeline.
- Applied SHAP values to interpret black-box decisions, identifying recent payment history as the most critical default predictor and translating the explanation into business-facing insight.

Agricultural Insurance Purchase Modeling

Econometric Modeling in R

UC Santa Cruz

GitHub Repository

- Reached 76% predictive precision by cleaning and merging Harvard Dataverse survey data (1,000+ records, 30+ variables) and comparing Logit baseline, Lasso feature selection, and Random Forest models in R.
- Interpreted model outputs through feature importance metrics to identify key behavioral drivers of insurance purchase and translated technical findings into actionable business insight.
- Ran error analysis on prediction variance and documented model limitations and survey biases, including the instability that rare purchase events introduce into classification.

DiD Analysis of COVID-19 Lockdown Policies

Causal Inference on Cross-Country Panel Data

UC Santa Cruz

GitHub Repository

- Quantified the causal impact of mandatory versus optional lockdown policies on infection and mortality rates per 100k across 5 major economies (US, China, Japan, UK, India) with a Difference-in-Differences framework in R.
- Aggregated and standardized cross-country economic panel data from the WHO and Our World in Data into a time-series panel suitable for quasi-experimental estimation.
- Reduced omitted-variable bias by incorporating control variables and validated the policy estimates through robustness checks against baseline specifications.